Definite Integration: Questions and Full Solutions

Question 01

English

Let $p : \mathbb{R} \to \mathbb{R}$ be a polynomial function of degree 3 such that $p(0) = 12$ and

$$\int_0^1 p(t)\,dt = \int_0^2 p(t)\,dt = \int_0^3 p(t)\,dt = 0.$$

Then $\displaystyle\int_4^5 p(t)\,dt$ is equal to

(A) 192
(B) -96
(C) -192
(D) -48

Only one option is correct.

हिन्दी

माना कि $p : \mathbb{R} \to \mathbb{R}$ घात 3 का एक बहुपद फलन है जिसके लिए $p(0) = 12$ तथा

$$\int_0^1 p(t)\,dt = \int_0^2 p(t)\,dt = \int_0^3 p(t)\,dt = 0.$$

तब $\displaystyle\int_4^5 p(t)\,dt$ बराबर है

(A) 192
(B) -96
(C) -192
(D) -48

केवल एक विकल्प सही है।

Solution

Define $P(x) = \int_0^x p(t)\,dt$. Since p is continuous, the first part of the fundamental theorem gives $P'(x) = p(x)$ for all x; as p is a cubic, P is a polynomial of degree 4.

$P(0) = 0$, and the three given integrals are $P(1)$, $P(2)$, $P(3)$, all equal to 0. A degree-4 polynomial with the four roots $0, 1, 2, 3$ is $P(x) = kx(x-1)(x-2)(x-3)$ for some constant $k \neq 0$.

Differentiate: $P'(0)$ equals k times the product of the other three factors at 0, i.e. $k(-1)(-2)(-3) = -6k$.

Since $P'(0) = p(0) = 12$, we get $-6k = 12$, so $k = -2$.

By the second part of the fundamental theorem, $\int_4^5 p(t)\,dt = P(5) - P(4)$.

$P(5) = k \cdot 5 \cdot 4 \cdot 3 \cdot 2 = 120k$ and $P(4) = k \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 24k$, so the integral is $96k = -192$.

Correct Option: (C)

Question 02

English

The value of the integral

$$\int_{\pi/2}^{3\pi/2} \frac{(x+\sin x)\,|\cos x|}{|\sin x|+|\cos x|}\,dx$$

is equal to

(A) $\dfrac{\pi^2}{2}$

(B) $\dfrac{\pi^2}{4}$

(C) π^2

(D) $\dfrac{\pi}{2}$

Only one option is correct.

हिन्दी

समाकल

$$\int_{\pi/2}^{3\pi/2} \frac{(x+\sin x)\,|\cos x|}{|\sin x|+|\cos x|}\,dx$$

का मान बराबर है

(A) $\dfrac{\pi^2}{2}$

(B) $\dfrac{\pi^2}{4}$

(C) π^2

(D) $\dfrac{\pi}{2}$

केवल एक विकल्प सही है।

Solution

Write $g(x) = \dfrac{|\cos x|}{|\sin x|+|\cos x|}$, so $I = \int_{\pi/2}^{3\pi/2}(x+\sin x)g(x)\,dx$.

Since $\sin(2\pi - x) = -\sin x$ and $\cos(2\pi - x) = \cos x$, we have $g(2\pi - x) = g(x)$.

Queen's rule on $[\pi/2, 3\pi/2]$ (here $a + b = 2\pi$): $I = \int_{\pi/2}^{3\pi/2}(2\pi - x - \sin x)g(x)\,dx$.

Adding the two forms of I: $2I = \int_{\pi/2}^{3\pi/2} 2\pi\, g(x)\,dx$, so $I = \pi \int_{\pi/2}^{3\pi/2} g(x)\,dx$.

The same conclusion through odd and even functions: put $x = \pi + t$; then $x + \sin x = \pi + (t - \sin t)$ and $g(\pi + t) = \dfrac{|\cos t|}{|\sin t|+|\cos t|}$ is even in t, while $(t - \sin t)$ is odd, so their product integrates to 0 over $[-\pi/2, \pi/2]$.

Now $\int_{\pi/2}^{3\pi/2} g(x)\,dx = \int_{-\pi/2}^{\pi/2} \dfrac{|\cos t|}{|\sin t|+|\cos t|}dt = 2\int_0^{\pi/2} \dfrac{\cos t}{\sin t+\cos t}dt$, because the integrand is even and on $[0, \pi/2]$ both $\sin t, \cos t \geq 0$.

Let $J = \int_0^{\pi/2} \dfrac{\cos t}{\sin t+\cos t}dt$. King's rule ($t \to \pi/2 - t$) gives $J = \int_0^{\pi/2} \dfrac{\sin t}{\cos t+\sin t}dt$; adding, $2J = \int_0^{\pi/2} 1\,dt = \dfrac{\pi}{2}$, so $J = \dfrac{\pi}{4}$.

Hence $\int_{\pi/2}^{3\pi/2} g = \dfrac{\pi}{2}$ and $I = \pi \cdot \dfrac{\pi}{2} = \dfrac{\pi^2}{2}$.

Correct Option: (A)

Question 03

English

Let $f : [0, 3] \to \mathbb{R}$ be defined by $f(x) = x^{[x]}$ for $x \neq 0$ and $f(0) = 1$, where $[t]$ denotes the greatest integer less than or equal to t. Then $\displaystyle\int_0^3 f(x)\,dx$ is equal to

(A) $\dfrac{47}{6}$
(B) $\dfrac{53}{6}$
(C) $\dfrac{26}{3}$
(D) 9

Only one option is correct.

हिन्दी

माना कि $f : [0, 3] \to \mathbb{R}$, $x \neq 0$ के लिए $f(x) = x^{[x]}$ तथा $f(0) = 1$ द्वारा परिभाषित है, जहाँ $[t]$, t से कम या उसके बराबर महत्तम पूर्णांक को दर्शाता है। तब $\displaystyle\int_0^3 f(x)\,dx$ बराबर है

(A) $\dfrac{47}{6}$
(B) $\dfrac{53}{6}$
(C) $\dfrac{26}{3}$
(D) 9

केवल एक विकल्प सही है।

Solution

The greatest integer $[x]$ takes the values $0, 1, 2$ on $[0, 1)$, $[1, 2)$, $[2, 3)$ respectively, so f is a power function on each of these subintervals; f jumps at $x = 1$ and $x = 2$ (from 1 to 1 at $x = 1$, from 2 to 4 at $x = 2$), and finitely many jumps do not affect the integral.

By additivity, $\int_0^3 f = \int_0^1 f + \int_1^2 f + \int_2^3 f$.

On $[0, 1)$: $f(x) = x^0 = 1$ (and $f(0) = 1$ by definition), so $\int_0^1 f = 1$.

On $[1, 2)$: $f(x) = x$, so $\int_1^2 x\,dx = \dfrac{2^2 - 1^2}{2} = \dfrac{3}{2}$.

On $[2, 3)$: $f(x) = x^2$, so $\int_2^3 x^2 dx = \dfrac{27 - 8}{3} = \dfrac{19}{3}$.

Adding: $1 + \dfrac{3}{2} + \dfrac{19}{3} = \dfrac{6 + 9 + 38}{6} = \dfrac{53}{6}$.

Correct Option: (B)

Question 04

English

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x - \dfrac{\sin(\pi x)}{2\pi}$. Then f is a bijection; let f^{-1} denote its inverse. Then the area of the region bounded by the curves $y = f(x)$ and $y = f^{-1}(x)$ and lying in the strip $0 \le x \le 3$ is

(A) $\dfrac{3}{\pi^2}$

(B) $\dfrac{3}{\pi}$

(C) $\dfrac{6}{\pi}$

(D) $\dfrac{6}{\pi^2}$

Only one option is correct.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x - \dfrac{\sin(\pi x)}{2\pi}$ द्वारा परिभाषित है। तब f एकैकी आच्छादक है; माना f^{-1} इसका प्रतिलोम फलन है। तब वक्रों $y = f(x)$ तथा $y = f^{-1}(x)$ से परिबद्ध तथा पट्टी $0 \le x \le 3$ में स्थित क्षेत्र का क्षेत्रफल है

(A) $\dfrac{3}{\pi^2}$

(B) $\dfrac{3}{\pi}$

(C) $\dfrac{6}{\pi}$

(D) $\dfrac{6}{\pi^2}$

केवल एक विकल्प सही है।

Solution

$f'(x) = 1 - \frac{1}{2}\cos(\pi x) \ge \frac{1}{2} > 0$, so f is strictly increasing and continuous; since $f(x) - x$ is bounded, $f(x) \to \pm\infty$ as $x \to \pm\infty$, so f is a bijection and f^{-1} exists and is increasing.

The graph of f^{-1} is the reflection of the graph of f in $y = x$. For an increasing function, $f(x) = f^{-1}(x)$ forces $f(x) = x$; here $f(x) = x \iff \sin(\pi x) = 0 \iff x \in \mathbb{Z}$. In the strip the curves meet at $(0,0), (1,1), (2,2), (3,3)$.

Between consecutive integers one curve lies above $y = x$ and the other below it, so the region is three closed loops, each symmetric about $y = x$; the area of a loop is twice the area between $y = f(x)$ and $y = x$.

On $[0,1]$: $x - f(x) = \dfrac{\sin \pi x}{2\pi} \ge 0$, so the loop has area $2\int_0^1 \dfrac{\sin \pi x}{2\pi} dx = \dfrac{1}{\pi} \cdot \dfrac{2}{\pi} = \dfrac{2}{\pi^2}$.

The same follows from the formula for a function and its inverse: $\int_0^1 f(x)dx + \int_{f(0)}^{f(1)} f^{-1}(y)dy = 1 \cdot f(1) - 0 \cdot f(0) = 1$. With $\int_0^1 f = \dfrac{1}{2} - \dfrac{1}{\pi^2}$ we get $\int_0^1 f^{-1} = \dfrac{1}{2} + \dfrac{1}{\pi^2}$, and the loop area is $\int_0^1 (f^{-1} - f) = \dfrac{2}{\pi^2}$.

On $[1,2]$ and $[2,3]$ the same computation with $|\sin \pi x|$ gives $\dfrac{2}{\pi^2}$ each.

Total area $= \dfrac{6}{\pi^2}$.

Correct Option: (D)

Question 05

English

Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function such that

$$f(x) = x^2 + \int_0^x e^{x-t} f(t)\, dt \quad \text{for all } x \in \mathbb{R}.$$

Then $f(1)$ is equal to

(A) $\dfrac{e^2 - 1}{4}$

(B) $\dfrac{e^2 + 1}{4}$

(C) $\dfrac{e^2 - 1}{2}$

(D) $\dfrac{e^2 - 3}{4}$

Only one option is correct.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$ एक सतत फलन है जिसके लिए

$$f(x) = x^2 + \int_0^x e^{x-t} f(t)\, dt, \quad x \in \mathbb{R}.$$

तब $f(1)$ बराबर है

(A) $\dfrac{e^2 - 1}{4}$

(B) $\dfrac{e^2 + 1}{4}$

(C) $\dfrac{e^2 - 1}{2}$

(D) $\dfrac{e^2 - 3}{4}$

केवल एक विकल्प सही है।

Solution

Take e^x out of the integral: $f(x) = x^2 + e^x \int_0^x e^{-t} f(t)\, dt$. Putting $x = 0$ gives $f(0) = 0$.

The right side is differentiable (product of e^x and an integral of a continuous function), so f is differentiable.

By the Newton-Leibniz rule, $\dfrac{d}{dx} \int_0^x e^{-t} f(t)\, dt = e^{-x} f(x)$. Hence $f'(x) = 2x + e^x \int_0^x e^{-t} f(t) dt + e^x \cdot e^{-x} f(x)$.

From the given equation, $e^x \int_0^x e^{-t} f(t) dt = f(x) - x^2$. So $f'(x) = 2x + f(x) - x^2 + f(x)$, i.e. $f' - 2f = 2x - x^2$.

This is a linear differential equation. Try $f_p = ax^2 + bx + c$: $2ax + b - 2ax^2 - 2bx - 2c = -x^2 + 2x$ gives $a = \frac{1}{2}$, $b = -\frac{1}{2}$, $c = -\frac{1}{4}$.

General solution $f(x) = \dfrac{x^2}{2} - \dfrac{x}{2} - \dfrac{1}{4} + Ce^{2x}$; $f(0) = 0$ gives $C = \dfrac{1}{4}$.

Check: this f satisfies the original integral equation (verified by direct integration).

$f(1) = \dfrac{1}{2} - \dfrac{1}{2} - \dfrac{1}{4} + \dfrac{e^2}{4} = \dfrac{e^2 - 1}{4}$.

Correct Option: (A)

Question 06

English

Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable function such that $f(x + y) = f(x) + f(y) + xy(x + y)$ for all $x, y \in \mathbb{R}$ and $\lim\limits_{x\to 0} \dfrac{f(x)}{\sin x} = -3$. Let $G : \mathbb{R} \to \mathbb{R}$ be defined by

$$G(x) = \int_x^{2x} f(t)\, dt.$$

Then the minimum value of G is

(A) $-\dfrac{81}{40}$

(B) $-\dfrac{81}{20}$

(C) $-\dfrac{27}{4}$

(D) $-\dfrac{9}{5}$

Only one option is correct.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$ एक अवकलनीय फलन है जिसके लिए सभी $x, y \in \mathbb{R}$ के लिए $f(x+y) = f(x)+f(y)+xy(x+y)$ तथा $\lim\limits_{x\to 0} \dfrac{f(x)}{\sin x} = -3$ है। माना कि $G : \mathbb{R} \to \mathbb{R}$

$$G(x) = \int_x^{2x} f(t)\, dt$$

द्वारा परिभाषित है। तब G का न्यूनतम मान है

(A) $-\dfrac{81}{40}$

(B) $-\dfrac{81}{20}$

(C) $-\dfrac{27}{4}$

(D) $-\dfrac{9}{5}$

केवल एक विकल्प सही है।

Solution

Put $x = y = 0$: $f(0) = 2f(0)$, so $f(0) = 0$.

Then $\dfrac{f(x)}{\sin x} = \dfrac{f(x) - f(0)}{x} \cdot \dfrac{x}{\sin x} \to f'(0)$, so $f'(0) = -3$.

For any x: $\dfrac{f(x + h) - f(x)}{h} = \dfrac{f(h)}{h} + x^2 + xh \to -3 + x^2$ as $h \to 0$. So $f'(x) = x^2 - 3$ and, with $f(0) = 0$, $f(x) = \dfrac{x^3}{3} - 3x$.

By the Newton-Leibniz rule, $G'(x) = f(2x) \cdot 2 - f(x) \cdot 1 = 2\left(\dfrac{8x^3}{3} - 6x\right) - \dfrac{x^3}{3} + 3x = 5x^3 - 9x = x(5x^2 - 9)$.

Critical points: $x = 0$ and $x = \pm\dfrac{3}{\sqrt{5}}$. Sign of G': negative on $\left(-\infty, -\frac{3}{\sqrt{5}}\right)$, positive on $\left(-\frac{3}{\sqrt{5}}, 0\right)$, negative on $\left(0, \frac{3}{\sqrt{5}}\right)$, positive after. So $x = \pm\frac{3}{\sqrt{5}}$ are local minima and $x = 0$ a local maximum.

Explicitly $G(x) = \left[\dfrac{t^4}{12} - \dfrac{3t^2}{2}\right]_x^{2x} = \dfrac{15x^4}{12} - \dfrac{9x^2}{2} = \dfrac{5}{4}x^4 - \dfrac{9}{2}x^2$, which tends to $+\infty$ as $|x| \to \infty$, so the least of the local minima is the minimum value.
At $x^2 = \dfrac{9}{5}$: $G = \dfrac{5}{4}\cdot\dfrac{81}{25} - \dfrac{9}{2}\cdot\dfrac{9}{5} = \dfrac{81}{20} - \dfrac{162}{20} = -\dfrac{81}{20}$.

Correct Option: (B)

Question 07

English

For each non-negative integer n, let $I_n = \displaystyle\int_0^\infty x^n e^{-4x}\,dx$. Then the value of $\displaystyle\sum_{n=0}^{\infty} \frac{2^n I_n}{n!}$ is

(A) $\dfrac{1}{4}$
(B) 1
(C) 2
(D) $\dfrac{1}{2}$

Only one option is correct.

हिन्दी

प्रत्येक अऋणात्मक पूर्णांक n के लिए माना $I_n = \displaystyle\int_0^\infty x^n e^{-4x}\,dx$ है। तब $\displaystyle\sum_{n=0}^{\infty} \frac{2^n I_n}{n!}$ का मान है

(A) $\dfrac{1}{4}$
(B) 1
(C) 2
(D) $\dfrac{1}{2}$

केवल एक विकल्प सही है।

Solution

With $t = 4x$: $I_n = \dfrac{1}{4^{n+1}}\int_0^\infty t^n e^{-t}dt = \dfrac{\Gamma(n+1)}{4^{n+1}} = \dfrac{n!}{4^{n+1}}$ (Gamma function, $\Gamma(n+1) = n!$).
So the general term is $\dfrac{2^n}{n!}\cdot\dfrac{n!}{4^{n+1}} = \dfrac{1}{4}\left(\dfrac{1}{2}\right)^n$.
This is a geometric series with first term $\dfrac{1}{4}$ and ratio $\dfrac{1}{2}$, so its sum is $\dfrac{1/4}{1-1/2} = \dfrac{1}{2}$.
Check: $\sum \frac{(2x)^n}{n!} = e^{2x}$, and $\int_0^\infty e^{2x}e^{-4x}dx = \frac{1}{2}$.

Correct Option: (D)

Question 08

English

Let $f : (0, \infty) \to \mathbb{R}$ be a function, differentiable at $x = 1$, such that $f(xy) = f(x) + f(y)$ for all $x, y \in (0, \infty)$ and $\lim\limits_{h \to 0} \dfrac{f(1+h)}{h} = 2$. Then

$$\lim_{x \to 1} \frac{1}{(x-1)^2} \int_x^{x^2} f(t)\, dt$$

is equal to

(A) 6
(B) 2
(C) 3
(D) $\dfrac{3}{2}$

Only one option is correct.

हिन्दी

माना कि $f : (0, \infty) \to \mathbb{R}$ एक फलन है जो $x = 1$ पर अवकलनीय है, जिसके लिए सभी $x, y \in (0, \infty)$ के लिए $f(xy) = f(x) + f(y)$ तथा $\lim\limits_{h \to 0} \dfrac{f(1+h)}{h} = 2$ है। तब

$$\lim_{x \to 1} \frac{1}{(x-1)^2} \int_x^{x^2} f(t)\, dt$$

बराबर है

(A) 6
(B) 2
(C) 3
(D) $\dfrac{3}{2}$

केवल एक विकल्प सही है।

Solution

Putting $x = y = 1$ gives $f(1) = 0$, so the given limit is $f'(1) = \lim_{h \to 0} \dfrac{f(1+h) - f(1)}{h} = 2$.

For $x > 0$: $f(x+h) = f\big(x\,(1 + \tfrac{h}{x})\big) = f(x) + f\big(1 + \tfrac{h}{x}\big)$, so $\dfrac{f(x+h) - f(x)}{h} = \dfrac{1}{x} \cdot \dfrac{f(1 + h/x)}{h/x} \to \dfrac{2}{x}$.

Hence $f'(x) = \dfrac{2}{x}$ and, with $f(1) = 0$, $f(x) = 2 \ln x$.

At $x = 1$ both $\int_x^{x^2} f$ and $(x-1)^2$ vanish; f is continuous, so L'Hospital's rule applies with the Newton-Leibniz rule.

$\dfrac{d}{dx} \int_x^{x^2} f(t)dt = f(x^2) \cdot 2x - f(x) = 2x \cdot 4 \ln x - 2 \ln x = (8x - 2) \ln x$, and $\dfrac{d}{dx}(x-1)^2 = 2(x-1)$.

So the limit equals $\lim_{x \to 1} \dfrac{(8x-2)}{2} \cdot \dfrac{\ln x}{x-1} = \dfrac{6}{2} \cdot 1 = 3$.

Correct Option: (C)

Question 09

English

Let $f : \mathbb{R} \to \mathbb{R}$ be a polynomial function such that

$$\int_{x-1}^{x+1} f(t)\,dt = x^3 \quad \text{for all } x \in \mathbb{R}.$$

Then the value of $f(-4)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$ एक बहुपद फलन है जिसके लिए

$$\int_{x-1}^{x+1} f(t)\,dt = x^3, \quad x \in \mathbb{R}.$$

तब $f(-4)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Differentiate both sides with the Newton-Leibniz rule (both limits depend on x): $f(x+1)\cdot 1 - f(x-1)\cdot 1 = 3x^2$ for all x.

If f has degree $m \geq 1$ with leading coefficient a, then $f(x+1) - f(x-1)$ has degree $m-1$ with leading coefficient $2ma$. Matching with $3x^2$: $m = 3$ and $6a = 3$, so $a = \frac{1}{2}$.

Write $f(x) = \frac{1}{2}x^3 + bx^2 + cx + d$. Then $f(x+1) - f(x-1) = \frac{1}{2}(6x^2+2) + b(4x) + 2c = 3x^2 + 4bx + 1 + 2c$.

Comparing with $3x^2$: $b = 0$ and $1 + 2c = 0$, so $c = -\frac{1}{2}$.

The derivative condition loses the constant d. Use the given equation itself at $x = 0$ (the integral evaluated by the second part of the fundamental theorem): $\int_{-1}^{1} \left(\frac{t^3}{2} - \frac{t}{2} + d\right) dt = 0 + 0 + 2d$, which must equal $0^3 = 0$. So $d = 0$.

Check: $\int_{x-1}^{x+1} \frac{t^3 - t}{2} dt = \left[\frac{t^4}{8} - \frac{t^2}{4}\right]_{x-1}^{x+1} = x^3 + x - x = x^3$.

Hence $f(x) = \dfrac{x^3 - x}{2}$ and $f(-4) = \dfrac{-64 + 4}{2} = -30$.

Final Answer: -30

Question 10

English

Let k be the positive integer for which

$$\int_0^{k\pi/2} \frac{dx}{\sin^2 x + 9\cos^2 x} = \frac{7\pi}{2}.$$

Then the value of k is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि k वह धनात्मक पूर्णांक है जिसके लिए

$$\int_0^{k\pi/2} \frac{dx}{\sin^2 x + 9\cos^2 x} = \frac{7\pi}{2}.$$

तब k का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $g(x) = \dfrac{1}{\sin^2 x + 9\cos^2 x}$, continuous and positive on $\mathbb{R}$. Dividing by $\cos^2 x$: $g(x) = \dfrac{\sec^2 x}{\tan^2 x + 9}$, so on each interval where $\cos x \neq 0$ an antiderivative is $F(x) = \dfrac{1}{3}\tan^{-1}\left(\dfrac{\tan x}{3}\right)$.

F is not continuous on $\mathbb{R}$: as $x \to \frac{\pi}{2}^-$, $F \to \frac{\pi}{6}$, while as $x \to \frac{\pi}{2}^+$, $F \to -\frac{\pi}{6}$. So $F(b) - F(a)$ is valid only when $[a, b]$ contains no point $\frac{\pi}{2} + m\pi$.

On $[0, \frac{\pi}{2}]$: $\int_0^{\pi/2} g = \lim_{x\to\pi/2^-} F(x) - F(0) = \dfrac{\pi}{6}$.

Since $g(\pi - x) = g(x)$, $\int_{\pi/2}^{\pi} g = \int_0^{\pi/2} g = \dfrac{\pi}{6}$; since $g(x + \pi) = g(x)$, the pattern repeats. So every interval $\left[\frac{m\pi}{2}, \frac{(m+1)\pi}{2}\right]$ contributes $\dfrac{\pi}{6}$, and $\int_0^{k\pi/2} g = \dfrac{k\pi}{6}$.

Setting $\dfrac{k\pi}{6} = \dfrac{7\pi}{2}$ gives $k = 21$.

Using $F\left(\frac{k\pi}{2}\right) - F(0)$ directly would give 0 for even k and $\pm\frac{\pi}{6}$ for odd k, which is wrong because F jumps by $-\frac{\pi}{3}$ at each odd multiple of $\frac{\pi}{2}$.

Final Answer: 21

Question 11

English

Let $F : \mathbb{R} \to \mathbb{R}$ be defined by

$$F(x) = \int_0^x \frac{1 + t^2}{1 - t^2 + t^4}\, dt.$$

Then the value of $F\left(-\dfrac{1+\sqrt{5}}{2}\right)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $F : \mathbb{R} \to \mathbb{R}$

$$F(x) = \int_0^x \frac{1 + t^2}{1 - t^2 + t^4}\, dt$$

द्वारा परिभाषित है। तब $F\left(-\dfrac{1+\sqrt{5}}{2}\right)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$1 - t^2 + t^4 = \left(t^2 - \frac{1}{2}\right)^2 + \frac{3}{4} > 0$, so the integrand is continuous on $\mathbb{R}$. By the first part of the

fundamental theorem, F is differentiable with $F'(x) = \dfrac{1+x^2}{1-x^2+x^4}$; F is odd since the integrand is even.

For $x \neq \pm 1$, $\dfrac{d}{dx}\tan^{-1}\dfrac{x}{1-x^2} = \dfrac{1+x^2}{(1-x^2)^2+x^2} = \dfrac{1+x^2}{1-x^2+x^4}$. So $\phi(x) = \tan^{-1}\dfrac{x}{1-x^2}$ is an antiderivative on each of $(-\infty,-1)$, $(-1,1)$, $(1,\infty)$, but not across ± 1.

On $(-1,1)$, F and ϕ have the same derivative and $F(0) = \phi(0) = 0$, so $F = \phi$ there. As $x \to -1^+$, $\frac{x}{1-x^2} \to -\infty$, so $F(-1) = -\frac{\pi}{2}$ (by continuity of F).

On $(-\infty,-1)$, $F = \phi + C$. As $x \to -1^-$, $\frac{x}{1-x^2} \to +\infty$ and $\phi \to \frac{\pi}{2}$. Continuity of F at -1 gives $\frac{\pi}{2} + C = -\frac{\pi}{2}$, so $C = -\pi$.

Let $x_0 = -\frac{1+\sqrt{5}}{2} < -1$. Then x_0 is a root of $x^2 + x - 1 = 0$, i.e. $x_0^2 + x_0 - 1 = 0$, so $1 - x_0^2 = x_0$ and $\frac{x_0}{1-x_0^2} = 1$.

Hence $F(x_0) = \tan^{-1} 1 - \pi = \frac{\pi}{4} - \pi = -\frac{3\pi}{4} = -2.356\ldots$, which is -2.36 to two decimal places. (The naive value $\phi(x_0) - \phi(0) = \frac{\pi}{4}$ ignores the jump.)

Cross-check: $\frac{1+t^2}{1-t^2+t^4} = \frac{1}{2}\left[\frac{1}{t^2+\sqrt{3}t+1} + \frac{1}{t^2-\sqrt{3}t+1}\right]$ gives the continuous antiderivative $\tan^{-1}(2x + \sqrt{3}) + \tan^{-1}(2x - \sqrt{3})$, whose value at x_0 is also $-\frac{3\pi}{4}$.

Final Answer: -2.36

Question 12

English

The value of $\displaystyle\int_{-1}^{3} \frac{|x|}{(1+x^2)^2}\,dx$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$\displaystyle\int_{-1}^{3} \frac{|x|}{(1+x^2)^2}\,dx$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The natural substitution is $u = x^2$, but on $[-1,3]$ the map $x \mapsto x^2$ is not monotonic (it decreases on $[-1,0]$ and increases on $[0,3]$), and $|x|\,dx$ equals $\frac{1}{2}du$ only where $x \geq 0$. So split at 0.

On $[-1,0]$: $|x| = -x$. With $u = x^2$, $du = 2x\,dx$, so $-x\,dx = -\frac{1}{2}du$; $x = -1 \mapsto u = 1$, $x = 0 \mapsto u = 0$.

Hence $\int_{-1}^{0} \frac{-x}{(1+x^2)^2}dx = -\frac{1}{2}\int_1^0 \frac{du}{(1+u)^2} = \frac{1}{2}\int_0^1 \frac{du}{(1+u)^2} = \frac{1}{2}\left(1 - \frac{1}{2}\right) = \frac{1}{4}$.

On $[0,3]$: $|x| = x$, u runs from 0 to 9: $\int_0^3 \frac{x\,dx}{(1+x^2)^2} = \frac{1}{2}\int_0^9 \frac{du}{(1+u)^2} = \frac{1}{2}\left(1 - \frac{1}{10}\right) = \frac{9}{20}$.

Total: $\dfrac{1}{4} + \dfrac{9}{20} = \dfrac{14}{20} = 0.70$.

A single substitution $u = x^2$ from $u = 1$ to $u = 9$ would give $\frac{1}{2}\left(\frac{1}{2} - \frac{1}{10}\right) = 0.20$, which is wrong because it treats $|x|\,dx$ as $\frac{1}{2}du$ on $[-1,0]$ too.

Final Answer: 0.70

Question 13

English

The value of $\displaystyle\int_1^{e^{12}} \frac{x \log_e x \, (12 - \log_e x)}{x^2 + e^{12}} \, dx$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$\displaystyle\int_1^{e^{12}} \frac{x \log_e x \, (12 - \log_e x)}{x^2 + e^{12}} \, dx$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Put $x = e^t$. On $[0, 12]$ this is increasing, so $x = 1 \mapsto t = 0$, $x = e^{12} \mapsto t = 12$, and $dx = e^t dt$, $\log_e x = t$.

Then $\dfrac{x\,dx}{x^2 + e^{12}} = \dfrac{e^{2t} dt}{e^{2t} + e^{12}} = \dfrac{dt}{1 + e^{12-2t}}$, so $I = \int_0^{12} \dfrac{t(12-t)}{1 + e^{12-2t}} dt$.

Let $h(t) = \dfrac{1}{1 + e^{12-2t}}$. Then $h(12 - t) = \dfrac{1}{1 + e^{2t-12}} = \dfrac{e^{12-2t}}{e^{12-2t} + 1} = 1 - h(t)$, while $t(12 - t)$ is unchanged by $t \to 12 - t$.

King's rule: $I = \int_0^{12} t(12 - t)\,(1 - h(t))\,dt$. Adding to the original form: $2I = \int_0^{12} t(12 - t)\,dt = 6 \cdot 144 - \dfrac{1728}{3} = 864 - 576 = 288$.

So $I = 144$.

Final Answer: 144

Question 14

English

Let $f : [0, 4] \to \mathbb{R}$ be defined by

$$f(x) = \frac{(x^2 - 4x + 5)\sqrt{x}}{\sqrt{x} + \sqrt{4 - x}}.$$

Let $I_1 = \displaystyle\int_0^4 f(x)\,dx$ and $I_2 = \int_1^3 f(x)\,dx$. Then the value of $\dfrac{I_1}{I_2}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : [0, 4] \to \mathbb{R}$

$$f(x) = \frac{(x^2 - 4x + 5)\sqrt{x}}{\sqrt{x} + \sqrt{4 - x}}$$

द्वारा परिभाषित है। माना $I_1 = \displaystyle\int_0^4 f(x)\,dx$ तथा $I_2 = \int_1^3 f(x)\,dx$ है। तब $\dfrac{I_1}{I_2}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Write $f = w\,h$ with $w(x) = x^2 - 4x + 5 = (x - 2)^2 + 1$ and $h(x) = \dfrac{\sqrt{x}}{\sqrt{x} + \sqrt{4 - x}}$.

Under $x \to 4-x$: $w(4-x) = (2-x)^2 + 1 = w(x)$ and $h(4-x) = \dfrac{\sqrt{4-x}}{\sqrt{4-x}+\sqrt{x}} = 1 - h(x)$.

King's rule on $[0,4]$: $I_1 = \int_0^4 w(4-x)h(4-x)dx = \int_0^4 w(x)(1-h(x))dx$. Adding: $2I_1 = \int_0^4 w = \left[\dfrac{(x-2)^3}{3} + x\right]_0^4 = \dfrac{16}{3} + 4 = \dfrac{28}{3}$, so $I_1 = \dfrac{14}{3}$.

Queen's rule on $[1,3]$ (here $a+b=4$, so the same map $x \to 4-x$): $2I_2 = \int_1^3 w = \left[\dfrac{(x-2)^3}{3} + x\right]_1^3 = \dfrac{2}{3} + 2 = \dfrac{8}{3}$, so $I_2 = \dfrac{4}{3}$.

Therefore $\dfrac{I_1}{I_2} = \dfrac{14/3}{4/3} = 3.50$.

Final Answer: 3.50

Question 15

English

The value of

$$\int_{-\pi/2}^{3\pi/2} \frac{\sin^4 x}{\left(\sin^4 x + \cos^4 x\right)\left(1 + e^{\cos x}\right)}\, dx$$

is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$$\int_{-\pi/2}^{3\pi/2} \frac{\sin^4 x}{\left(\sin^4 x + \cos^4 x\right)\left(1 + e^{\cos x}\right)}\, dx$$

का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $g(x) = \dfrac{\sin^4 x}{\sin^4 x + \cos^4 x}$ and $p(x) = \dfrac{1}{1+e^{\cos x}}$, so $I = \int_{-\pi/2}^{3\pi/2} g\,p\,dx$.

Queen's rule on $[-\frac{\pi}{2}, \frac{3\pi}{2}]$ (here $a+b=\pi$): replace x by $\pi - x$. Since $\sin(\pi-x) = \sin x$ and $\cos(\pi-x) = -\cos x$, $g(\pi-x) = g(x)$ and $p(\pi-x) = \dfrac{1}{1+e^{-\cos x}} = \dfrac{e^{\cos x}}{e^{\cos x}+1} = 1 - p(x)$.

So $I = \int g(1-p)\,dx$ and, adding, $2I = \int_{-\pi/2}^{3\pi/2} g(x)\,dx$.

Split $[-\frac{\pi}{2}, \frac{3\pi}{2}]$ at 0 and π. Even function: $g(-x) = g(x)$, so $\int_{-\pi/2}^{0} g = \int_0^{\pi/2} g$.

Property $\int_0^{2a} g = 2\int_0^a g$ when $g(2a-x) = g(x)$, with $a = \frac{\pi}{2}$: $\int_0^{\pi} g = 2\int_0^{\pi/2} g$.

Queen's rule on $[\pi, \frac{3\pi}{2}]$ via $x \to \pi - x$ maps it onto $[-\frac{\pi}{2}, 0]$: $\int_{\pi}^{3\pi/2} g = \int_{-\pi/2}^{0} g = \int_0^{\pi/2} g$.

Hence $\int_{-\pi/2}^{3\pi/2} g = 4\int_0^{\pi/2} g$.

King's rule on $[0, \frac{\pi}{2}]$: $g\left(\frac{\pi}{2} - x\right) = \dfrac{\cos^4 x}{\cos^4 x + \sin^4 x} = 1 - g(x)$, so $2\int_0^{\pi/2} g = \frac{\pi}{2}$ and $\int_0^{\pi/2} g = \frac{\pi}{4}$.

Therefore $2I = \pi$, $I = \dfrac{\pi}{2} = 1.5707\ldots$, which is 1.57 to two decimal places.

Final Answer: 1.57

Question 16

English

Considering the principal value of $\cot^{-1}$, which takes values in $(0, \pi)$, let

$$L = \lim_{n\to\infty} \sum_{k=1}^{2n} \frac{k(k-2n)}{n^3} \cot^{-1}\left(\frac{(k-n)^3}{n^3}\right).$$

Then the value of L is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$\cot^{-1}$ के मुख्य मान (जो $(0, \pi)$ में है) को लेते हुए माना

$$L = \lim_{n\to\infty} \sum_{k=1}^{2n} \frac{k(k-2n)}{n^3} \cot^{-1}\left(\frac{(k-n)^3}{n^3}\right).$$

तब L का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Put $x_k = \frac{k}{n}$, $\Delta x = \frac{1}{n}$; as k runs from 1 to $2n$, x_k runs through $(0, 2]$. The term is $\frac{1}{n} \cdot x_k(x_k - 2)\cot^{-1}\left((x_k - 1)^3\right)$, so by the definition of the integral as a limit of a sum,

$$L = \int_0^2 x(x-2)\cot^{-1}\left((x-1)^3\right) dx.$$

King's rule on $[0, 2]$ $(x \to 2-x)$: $x(x-2) \to (2-x)(-x) = x(x-2)$, and $(x-1)^3 \to (1-x)^3 = -(x-1)^3$ because $y \mapsto y^3$ is an odd function.

For the principal value, $\cot^{-1}(-y) = \pi - \cot^{-1} y$. So $L = \int_0^2 x(x-2)\left[\pi - \cot^{-1}\left((x-1)^3\right)\right] dx = \pi \int_0^2 x(x-2)dx - L$.

$\int_0^2 (x^2 - 2x)dx = \frac{8}{3} - 4 = -\frac{4}{3}$, so $2L = -\frac{4\pi}{3}$ and $L = -\frac{2\pi}{3} = -2.094\ldots$

To two decimal places, $L = -2.09$.

Final Answer: -2.09

Question 17

English

Let

$$\int_{\pi/8}^{3\pi/8} \frac{dx}{\sin^4 x\, \cos^4 x\, (1 + \tan^5 x)} = \frac{p}{q},$$

where p and q are positive integers with $\gcd(p, q) = 1$. Then the value of $p + q$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना

$$\int_{\pi/8}^{3\pi/8} \frac{dx}{\sin^4 x\,\cos^4 x\,(1+\tan^5 x)} = \frac{p}{q}$$

है, जहाँ p तथा q धनात्मक पूर्णांक हैं तथा $\gcd(p,q)=1$ है। तब $p+q$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let I be the integral. Queen's rule on $\left[\frac{\pi}{8}, \frac{3\pi}{8}\right]$ uses $x \to \frac{\pi}{8} + \frac{3\pi}{8} - x = \frac{\pi}{2} - x$.

Under this map $\sin x \leftrightarrow \cos x$, so $\sin^4 x \cos^4 x$ is unchanged and $\tan x \to \cot x$. Since $\dfrac{1}{1+\cot^5 x} = \dfrac{\tan^5 x}{\tan^5 x + 1}$, the two factors $\dfrac{1}{1+\tan^5 x}$ and $\dfrac{\tan^5 x}{1+\tan^5 x}$ add to 1.

Hence $2I = \int_{\pi/8}^{3\pi/8} \dfrac{dx}{\sin^4 x \cos^4 x}$. With $\sin x \cos x = \frac{1}{2}\sin 2x$, $\dfrac{1}{\sin^4 x \cos^4 x} = \dfrac{16}{\sin^4 2x} = 16\csc^4 2x$.

Put $u = 2x$ ($du = 2dx$, limits $\frac{\pi}{4}$ to $\frac{3\pi}{4}$): $2I = 8\int_{\pi/4}^{3\pi/4} \csc^4 u\,du$, so $I = 4\int_{\pi/4}^{3\pi/4} \csc^4 u\,du$.

$\csc^4 u = (1+\cot^2 u)\csc^2 u$; with $v = \cot u$, $dv = -\csc^2 u\,du$, an antiderivative is $-\cot u - \dfrac{\cot^3 u}{3}$ (continuous on $(0,\pi)$).

At $u = \frac{3\pi}{4}$, $\cot u = -1$: value $1 + \frac{1}{3} = \frac{4}{3}$. At $u = \frac{\pi}{4}$, $\cot u = 1$: value $-\frac{4}{3}$. Difference $\frac{8}{3}$.

So $I = 4 \cdot \frac{8}{3} = \dfrac{32}{3}$; $p = 32$, $q = 3$, $p + q = 35$.

Final Answer: 35

Question 18

English

Let $f : [-8, 8] \to \mathbb{R}$ be defined by $f(x) = x^5 \cos x + |x|\,(8-|x|)\,\big|4-|x|\big|$. Then the value of $\int_{-8}^{8} f(x)\,dx$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : [-8, 8] \to \mathbb{R}$, $f(x) = x^5 \cos x + |x|\,(8-|x|)\,\big|4-|x|\big|$ द्वारा परिभाषित है। तब $\int_{-8}^{8} f(x)\,dx$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Split $f = O + E$ with $O(x) = x^5 \cos x$ and $E(x) = |x|(8-|x|)\big|4-|x|\big|$.

$O(-x) = -x^5 \cos x = -O(x)$, so O is odd and $\int_{-8}^{8} O = 0$. E depends only on $|x|$, so it is even and $\int_{-8}^{8} E = 2\int_0^8 E = 2\int_0^8 x(8-x)|4-x|\,dx$.

Let $g(x) = x(8-x)|4-x|$ on $[0,8]$. Then $g(8-x) = (8-x)x|x-4| = g(x)$, so by the property $\int_0^{2a} g = 2\int_0^a g$ (with $a = 4$): $\int_0^8 g = 2\int_0^4 x(8-x)(4-x)\,dx$; on $[0,4]$ the modulus is simply $4-x$.

$x(8-x)(4-x) = 32x - 12x^2 + x^3$, and $\int_0^4 = 16 \cdot 16 - 4 \cdot 64 + \frac{256}{4} = 256 - 256 + 64 = 64$.

So $\int_{-8}^{8} f = 2 \cdot 2 \cdot 64 = 256$.

Final Answer: 256

Question 19

English

The value of $\displaystyle\int_0^4 \left(x^2-4x\right)\left|x^2-4x+3\right|\,dx$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$\displaystyle\int_0^4 \left(x^2-4x\right)\left|x^2-4x+3\right|\,dx$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $f(x)=(x^2-4x)|x^2-4x+3|$. Since $(4-x)^2-4(4-x)=x^2-4x$, we have $f(4-x)=f(x)$, so by the property $\int_0^{2a} f = 2\int_0^a f$ (with $a=2$), $\int_0^4 f = 2\int_0^2 f$.

Write $q=x^2-4x+3=(x-1)(x-3)$, so $x^2-4x=q-3$ and $f=(q-3)|q|$. On $[0,1]$, $q\ge 0$; on $[1,2]$, $q\le 0$.

So $\int_0^2 f=\int_0^1 (q^2-3q)\,dx-\int_1^2 (q^2-3q)\,dx$.

$\int_0^1 q\,dx=\frac{1}{3}-2+3=\frac{4}{3}$ and $\int_1^2 q\,dx=\left(\frac{8}{3}-8+6\right)-\frac{4}{3}=-\frac{2}{3}$.

$q^2=x^4-8x^3+22x^2-24x+9$: $\int_0^1 q^2=\frac{1}{5}-2+\frac{22}{3}-12+9=\frac{38}{15}$ and $\int_0^2 q^2=\frac{32}{5}-32+\frac{176}{3}-48+18=\frac{46}{15}$, so $\int_1^2 q^2=\frac{8}{15}$.

$\int_0^2 f=\left(\frac{38}{15}-4\right)-\left(\frac{8}{15}+2\right)=\frac{30}{15}-6=-4$.

Hence $\int_0^4 f=-8$.

Final Answer: -8

Question 20

English

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = \dfrac{\sin^2 x}{1 + \cos^2 x}$. Then the value of $\lim\limits_{T\to\infty} \dfrac{1}{T} \displaystyle\int_0^T f(x)\,dx$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \dfrac{\sin^2 x}{1 + \cos^2 x}$ द्वारा परिभाषित है। तब $\lim\limits_{T\to\infty} \dfrac{1}{T} \displaystyle\int_0^T f(x)\,dx$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$f(x + \pi) = f(x)$, so f is periodic with period π; let $A = \int_0^\pi f$. For $T > 0$ write $n = [T/\pi]$; since $f \geq 0$, $nA \leq \int_0^T f \leq (n+1)A$, and $\frac{n\pi}{T} \to 1$. Hence the limit equals $\dfrac{A}{\pi}$ (the average over one period).

Since $f(\pi - x) = f(x)$, the property $\int_0^{2a} f = 2\int_0^a f$ with $a = \frac{\pi}{2}$ gives $A = 2\int_0^{\pi/2} f$. Working on $[0, \frac{\pi}{2})$ also avoids the discontinuity of $\tan x$ at $\frac{\pi}{2}$.

$$f = \frac{2 - (1 + \cos^2 x)}{1 + \cos^2 x} = \frac{2}{1 + \cos^2 x} - 1.$$

$$\int_0^{\pi/2} \frac{dx}{1 + \cos^2 x} = \int_0^{\pi/2} \frac{\sec^2 x\,dx}{\tan^2 x + 2} = \int_0^\infty \frac{dt}{t^2 + 2} = \frac{1}{\sqrt{2}} \cdot \frac{\pi}{2}.$$

So $\int_0^{\pi/2} f = \dfrac{\pi}{\sqrt{2}} - \dfrac{\pi}{2}$ and $A = \pi\left(\sqrt{2} - 1\right)$.

The limit is $\dfrac{A}{\pi} = \sqrt{2} - 1 = 0.4142\ldots$, which is 0.41 to two decimal places.

Final Answer: 0.41

Question 21

English

Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function such that $f(x+1) = -f(x)$ for all $x \in \mathbb{R}$ and $\displaystyle\int_0^1 f(x)\,dx = 2$. Then the value of $\displaystyle\int_0^{20} x\,f(x)\,dx$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$ एक सतत फलन है जिसके लिए सभी $x \in \mathbb{R}$ के लिए $f(x+1) = -f(x)$ तथा $\displaystyle\int_0^1 f(x)\,dx = 2$ है। तब $\displaystyle\int_0^{20} x\,f(x)\,dx$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Applying the condition twice, $f(x+2) = -f(x+1) = f(x)$, so f is periodic with period 2; also $\int_1^2 f = \int_0^1 f(t+1)dt = -\int_0^1 f = -2$, so $\int_0^2 f = 0$.

Split $[0, 20]$ into the ten periods $[2m, 2m+2]$, $m = 0, \ldots, 9$. With $x = t + 2m$ and $f(t+2m) = f(t)$: $\int_{2m}^{2m+2} xf(x)\,dx = \int_0^2 (t+2m)f(t)\,dt = \int_0^2 tf(t)\,dt + 2m\int_0^2 f(t)\,dt = \int_0^2 tf(t)\,dt$.

Let $B = \int_0^1 tf(t)\,dt$. Then $\int_1^2 tf(t)\,dt = \int_0^1 (s+1)f(s+1)\,ds = -\int_0^1 (s+1)f(s)\,ds = -B - 2$.

So $\int_0^2 tf(t)\,dt = B - B - 2 = -2$, independent of the unknown B.

Hence $\int_0^{20} xf(x)\,dx = 10 \times (-2) = -20$. (Example: $f(x) = \pi \sin \pi x$ satisfies the conditions and gives -20.)

Final Answer: -20

Question 22

English

Let $f : \mathbb{R} \setminus \{1, 2\} \to \mathbb{R}$ be defined by $f(x) = \dfrac{x^3 - 3x^2 + 2x}{|x^2 - 3x + 2|}$. Then the value of $\displaystyle\int_{-3}^{2} f(x)\,dx$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \setminus \{1, 2\} \to \mathbb{R}$, $f(x) = \dfrac{x^3 - 3x^2 + 2x}{|x^2 - 3x + 2|}$ द्वारा परिभाषित है। तब $\displaystyle\int_{-3}^{2} f(x)\,dx$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Factor: $x^3 - 3x^2 + 2x = x(x^2 - 3x + 2) = x(x-1)(x-2)$. So for $x \neq 1, 2$, $f(x) = \dfrac{x(x-1)(x-2)}{|(x-1)(x-2)|}$.
$(x-1)(x-2) > 0$ for $x < 1$ and $x > 2$, and < 0 for $1 < x < 2$. Hence $f(x) = x$ on $[-3, 1)$ and $f(x) = -x$ on $(1, 2)$.
f jumps from 1 to -1 at $x = 1$ and is undefined at $1, 2$; these points do not affect the integral, so by additivity $\int_{-3}^{2} f = \int_{-3}^{1} x\,dx + \int_{1}^{2}(-x)\,dx$.
$\int_{-3}^{1} x\,dx = \frac{1-9}{2} = -4$ and $\int_{1}^{2}(-x)\,dx = -\frac{4-1}{2} = -\frac{3}{2}$.
Total $= -\frac{11}{2} = -5.50$.

Final Answer: -5.50

Question 23

English

Let $f : [0, 14] \to \mathbb{R}$ be defined by $f(x) = \max\{[x], |x - 7|\}$, where $[t]$ denotes the greatest integer less than or equal to t. Then the value of $\displaystyle\int_{0}^{14} f(x)\,dx$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : [0, 14] \to \mathbb{R}$, $f(x) = \max\{[x], |x - 7|\}$ द्वारा परिभाषित है, जहाँ $[t]$, t से कम या उसके बराबर महत्तम पूर्णांक को दर्शाता है। तब $\displaystyle\int_{0}^{14} f(x)\,dx$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

On each $[n, n+1)$, $n = 0, 1, \ldots, 13$, $[x] = n$, so $f(x) = \max\{n, |x - 7|\}$ there; by additivity the integral is the sum over these 14 subintervals.
For $0 \leq x < 4$: $|x - 7| = 7 - x > 3$, and $n \leq 3$, so $f(x) = 7 - x$. Contribution $\int_{0}^{4}(7 - x)\,dx = 28 - 8 = 20$.
For $4 \leq x < 7$: $|x - 7| = 7 - x \leq 3 < 4 \leq n$, so $f(x) = n$. Contribution $4 + 5 + 6 = 15$.
For $7 \leq x < 14$: $|x-7| = x-7 < n+1-7 = n-6 < n$, so $f(x) = n$. Contribution $7+8+\cdots+13 = 70$.
Total $= 20 + 15 + 70 = 105$.

Final Answer: 105

Question 24

English

For a real number t, let $\{t\} = t - [t]$, where $[t]$ denotes the greatest integer less than or equal to t. Then the value of $\displaystyle\int_{1/4}^{1} \left\{\frac{1}{x}\right\} dx$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

वास्तविक संख्या t के लिए माना $\{t\} = t - [t]$, जहाँ $[t]$, t से कम या उसके बराबर महत्तम पूर्णांक को दर्शाता है। तब $\displaystyle\int_{1/4}^{1} \left\{\frac{1}{x}\right\} dx$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For $x \in \left(\frac{1}{n+1}, \frac{1}{n}\right)$, $\frac{1}{x} \in (n, n+1)$, so $\left[\frac{1}{x}\right] = n$ and $\left\{\frac{1}{x}\right\} = \frac{1}{x} - n$. The breakpoints in $\left[\frac{1}{4}, 1\right]$ are $\frac{1}{3}$ and $\frac{1}{2}$.

By additivity, $\int_{1/4}^{1} = \int_{1/2}^{1} + \int_{1/3}^{1/2} + \int_{1/4}^{1/3}$, with $n = 1, 2, 3$ respectively.

$\int_{1/(n+1)}^{1/n} \left(\frac{1}{x} - n\right) dx = \ln\frac{n+1}{n} - n\left(\frac{1}{n} - \frac{1}{n+1}\right) = \ln\frac{n+1}{n} - \frac{1}{n+1}$.

Sum: $\left(\ln 2 + \ln\frac{3}{2} + \ln\frac{4}{3}\right) - \left(\frac{1}{2} + \frac{1}{3} + \frac{1}{4}\right) = \ln 4 - \frac{13}{12}$.

$\ln 4 = 1.38629\ldots$, so the value is $0.30296\ldots$, which is 0.30 to two decimal places.

Final Answer: 0.30

Question 25

English

Let $f : [0, 5] \to \mathbb{R}$ be defined by $f(x) = \min\left\{\{x\}, \dfrac{|x-3|}{3}\right\}$, where $\{t\} = t - [t]$ and $[t]$ denotes the greatest integer less than or equal to t. Then the value of $\displaystyle\int_0^5 f(x)\,dx$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : [0, 5] \to \mathbb{R}$, $f(x) = \min\left\{\{x\}, \dfrac{|x-3|}{3}\right\}$ द्वारा परिभाषित है, जहाँ $\{t\} = t - [t]$ तथा $[t]$, t से कम या उसके बराबर महत्तम पूर्णांक को दर्शाता है। तब $\displaystyle\int_0^5 f(x)\,dx$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

On each $[n, n+1)$, $\{x\} = x - n$, a line rising from 0 to 1. The other function is $g(x) = \frac{3-x}{3}$ for $x \le 3$ and $g(x) = \frac{x-3}{3}$ for $x \ge 3$. By additivity integrate over the five unit intervals, splitting each where $x - n = g(x)$.

$[0, 1)$: $x = \frac{3-x}{3} \iff x = \frac{3}{4}$. For $x < \frac{3}{4}$ the minimum is x, after it $1 - \frac{x}{3}$. Integral $= \frac{1}{2} \cdot \frac{9}{16} + \left[x - \frac{x^2}{6}\right]_{3/4}^{1} = \frac{9}{32} + \frac{17}{96} = \frac{11}{24}$.

$[1,2)$: $x-1=\frac{3-x}{3} \iff x=\frac{3}{2}$. Integral $=\frac{1}{2}\cdot\frac{1}{4}+\left[x-\frac{x^2}{6}\right]_{3/2}^{2}=\frac{1}{8}+\frac{5}{24}=\frac{1}{3}$.

$[2,3)$: $x-2=\frac{3-x}{3} \iff x=\frac{9}{4}$. Integral $=\frac{1}{2}\cdot\frac{1}{16}+\left[x-\frac{x^2}{6}\right]_{9/4}^{3}=\frac{1}{32}+\frac{3}{32}=\frac{1}{8}$.

$[3,4)$: $\frac{x-3}{3}\le x-3$ throughout, so the minimum is $\frac{x-3}{3}$; integral $=\frac{1}{6}$.

$[4,5)$: $x-4=\frac{x-3}{3} \iff x=\frac{9}{2}$. Integral $=\frac{1}{2}\cdot\frac{1}{4}+\left[\frac{(x-3)^2}{6}\right]_{9/2}^{5}=\frac{1}{8}+\frac{4-\frac{9}{4}}{6}=\frac{1}{8}+\frac{7}{24}=\frac{5}{12}$.

Sum: $\frac{11}{24}+\frac{8}{24}+\frac{3}{24}+\frac{4}{24}+\frac{10}{24}=\frac{36}{24}=1.50$.

Final Answer: 1.50

Question 26

English

Let $g:\mathbb{R}\to\mathbb{R}$ be the function such that $(g(x))^3+g(x)=x$ for all $x\in\mathbb{R}$. Then the value of $\int_2^{30} g(x)\,dx$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $g:\mathbb{R}\to\mathbb{R}$ वह फलन है जिसके लिए सभी $x\in\mathbb{R}$ के लिए $(g(x))^3+g(x)=x$ है। तब $\int_2^{30} g(x)\,dx$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $h(t)=t^3+t$. Since $h'(t)=3t^2+1>0$ and $h(t)\to\pm\infty$ as $t\to\pm\infty$, h is a bijection of $\mathbb{R}$. The condition $h(g(x))=x$ says $g=h^{-1}$ (so g exists, is unique and increasing).

$h(1)=2$ and $h(3)=30$, so $g(2)=1$ and $g(30)=3$.

For an increasing bijection, $\int_a^b h(t)\,dt+\int_{h(a)}^{h(b)} h^{-1}(y)\,dy = b\,h(b)-a\,h(a)$ (the two regions fill the rectangle difference). With $a=1$, $b=3$: $\int_1^3 h+\int_2^{30} g=3\cdot 30-1\cdot 2=88$.

$\int_1^3 (t^3+t)\,dt=\left[\frac{t^4}{4}+\frac{t^2}{2}\right]_1^3=20+4=24$.

So $\int_2^{30} g(x)\,dx=88-24=64$.

Final Answer: 64

Question 27

English

Let $F : \mathbb{R} \to \mathbb{R}$ be defined by $F(x) = \displaystyle\int_x^{x+1} t^2\, 2^{-t}\, dt$. If F has a local maximum at $x = \alpha$, then the value of α is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $F : \mathbb{R} \to \mathbb{R}$, $F(x) = \displaystyle\int_x^{x+1} t^2\, 2^{-t}\, dt$ द्वारा परिभाषित है। यदि $x = \alpha$ पर F का स्थानीय अधिकतम है, तो α का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The integrand $t^2 2^{-t}$ is continuous, so by the Newton-Leibniz rule $F'(x) = (x+1)^2 2^{-(x+1)} - x^2 2^{-x} = 2^{-x-1}\left[(x+1)^2 - 2x^2\right]$.

$(x+1)^2 - 2x^2 = -x^2 + 2x + 1 = -(x^2 - 2x - 1)$, which vanishes at $x = 1 \pm \sqrt{2}$.

Since $2^{-x-1} > 0$, the sign of F' is that of $-(x^2 - 2x - 1)$: negative for $x < 1 - \sqrt{2}$, positive for $1 - \sqrt{2} < x < 1 + \sqrt{2}$, negative for $x > 1 + \sqrt{2}$.

So F has a local minimum at $1 - \sqrt{2}$ and its only local maximum at $1 + \sqrt{2}$ (first derivative test). Hence $\alpha = 1 + \sqrt{2} = 2.414\ldots$, which is 2.41 to two decimal places.

Final Answer: 2.41

Question 28

English

Let $F : \mathbb{R} \to \mathbb{R}$ be defined by $F(x) = \displaystyle\int_{-2}^{x} (t-3)\,|t-2|\, dt$. Then the minimum value of F is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $F : \mathbb{R} \to \mathbb{R}$, $F(x) = \displaystyle\int_{-2}^{x} (t-3)\,|t-2|\, dt$ द्वारा परिभाषित है। तब F का न्यूनतम मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The integrand is continuous, so $F'(x) = (x-3)|x-2|$.

Critical points: $x = 2$ and $x = 3$. For $x < 3$, $x \neq 2$: $x - 3 < 0$ and $|x-2| > 0$, so $F' < 0$; for $x > 3$, $F' > 0$. Thus F is decreasing on $(-\infty, 3]$ (the point 2 is stationary but not an extremum) and increasing on $[3, \infty)$, and $F(3)$ is the minimum value.

$F(3) = \int_{-2}^{2}(t-3)(2-t)\, dt + \int_{2}^{3}(t-3)(t-2)\, dt$.

First: $(t-3)(2-t) = -t^2 + 5t - 6$; $\int_{-2}^{2} = -\frac{16}{3} + 0 - 24 = -\frac{88}{3}$.

Second: $\int_2^3 (t-2)(t-3)\, dt = -\frac{1}{6}$.

$F(3) = -\frac{88}{3} - \frac{1}{6} = -\frac{177}{6} = -29.50$.

Final Answer: -29.50

Question 29

English

Let $G : \mathbb{R} \to \mathbb{R}$ be defined by

$$G(x) = \int_0^{x^2+2x} (t+1)^2(t-3)\, e^{t^2}\, dt.$$

Then the sum of all the points of local extremum of G is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $G : \mathbb{R} \to \mathbb{R}$

$$G(x) = \int_0^{x^2+2x} (t+1)^2(t-3)\, e^{t^2}\, dt$$

द्वारा परिभाषित है। तब G के सभी स्थानीय चरम बिन्दुओं का योग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $u(x) = x^2 + 2x$. By the Newton-Leibniz rule, $G'(x) = u'(x)\,(u+1)^2(u-3)e^{u^2} = (2x+2)(u+1)^2(u-3)e^{u^2}$.

$u + 1 = x^2 + 2x + 1 = (x+1)^2$, so $(u+1)^2 = (x+1)^4$; $u - 3 = x^2 + 2x - 3 = (x+3)(x-1)$. Hence $G'(x) = 2(x+1)^5(x+3)(x-1)e^{u^2}$.

Critical points: $x = -3, -1, 1$. Since $e^{u^2} > 0$, the sign of G' is that of $(x+1)^5(x+3)(x-1)$.

$x < -3$: $(-)(-)(-) < 0$. $-3 < x < -1$: $(-)(+)(-) > 0$. $-1 < x < 1$: $(+)(+)(-) < 0$. $x > 1$: $(+)(+)(+) > 0$.

First derivative test: local minimum at $x = -3$, local maximum at $x = -1$, local minimum at $x = 1$. At $x = -1$ the factor $(x+1)^5$ gives $G''(-1) = 0$, so the second derivative test is inconclusive there; the first derivative test settles it.

Sum of the points of local extremum $= -3 + (-1) + 1 = -3$.

Final Answer: -3

Question 30

English

The value of $\lim\limits_{n\to\infty} \dfrac{1}{n^3} \displaystyle\sum_{k=1}^{6n} \left|k^2 - 9n^2\right|$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$\lim\limits_{n\to\infty} \dfrac{1}{n^3} \displaystyle\sum_{k=1}^{6n} \left|k^2 - 9n^2\right|$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Write $\frac{1}{n^3}\left|k^2-9n^2\right| = \frac{1}{n}\left|\left(\frac{k}{n}\right)^2 - 9\right|$. With $x_k = \frac{k}{n}$ and $\Delta x = \frac{1}{n}$, as k runs from 1 to $6n$, x_k runs through $(0, 6]$.

By the definition of the definite integral as a limit of a sum, the limit is $\int_0^6 |x^2-9|\,dx$ ($|x^2-9|$ is continuous).

$x^2 - 9 < 0$ on $[0, 3)$ and > 0 on $(3, 6]$: $\int_0^3 (9-x^2)dx = 27 - 9 = 18$ and $\int_3^6 (x^2-9)dx = (72-54) - (9-27) = 18 + 18 = 36$.

Limit $= 18 + 36 = 54$.

Final Answer: 54

Question 31

English

The value of $\lim\limits_{n\to\infty}\left(\prod\limits_{k=1}^{n}\frac{2k}{n+k}\right)^{1/n}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$\lim\limits_{n\to\infty}\left(\prod\limits_{k=1}^{n}\frac{2k}{n+k}\right)^{1/n}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $P_n = \left(\prod_{k=1}^{n}\frac{2k}{n+k}\right)^{1/n}$. Taking the natural logarithm, $\log_e P_n = \frac{1}{n}\sum_{k=1}^{n}\log_e\frac{2\,(k/n)}{1+k/n}$.

This is a Riemann sum for $f(x) = \log_e\frac{2x}{1+x}$ on $(0, 1]$. The singularity of $\log_e x$ at 0 is integrable and f is increasing, so the sum converges to $\int_0^1 f(x)\,dx$.

$\int_0^1 \log_e 2\,dx = \log_e 2$; $\int_0^1 \log_e x\,dx = [x\log_e x - x]_0^1 = -1$; $\int_0^1 \log_e(1+x)\,dx = [(1+x)\log_e(1+x) - x]_0^1 = 2\log_e 2 - 1$.

So $\lim \log_e P_n = \log_e 2 - 1 - (2\log_e 2 - 1) = -\log_e 2$.

By continuity of the exponential function, $\lim P_n = e^{-\log_e 2} = \frac{1}{2} = 0.50$.

Final Answer: 0.50

Question 32

English

Let $I = \displaystyle\int_0^{30} \frac{x}{1+e^{-x}}\,dx$. Then the greatest integer less than or equal to I is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $I = \displaystyle\int_0^{30} \frac{x}{1+e^{-x}}\,dx$ है। तब I से कम या उसके बराबर महत्तम पूर्णांक _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$\dfrac{x}{1+e^{-x}} = \dfrac{xe^x}{e^x+1} = x - \dfrac{x}{e^x+1}$, so $I = \int_0^{30} x\,dx - D = 450 - D$, where $D = \int_0^{30} \dfrac{x}{e^x+1}dx$.
For $x > 0$: $\dfrac{x}{e^x+1} > 0$ and $\dfrac{x}{e^x+1} < \dfrac{x}{e^x} = xe^{-x}$.
Hence $0 < D < \int_0^{30} xe^{-x}dx = [-(x+1)e^{-x}]_0^{30} = 1 - 31e^{-30} < 1$.
So $449 < I < 450$ and the greatest integer less than or equal to I is 449. (Numerically $I \approx 449.18$.)

Final Answer: 449

Question 33

English

Let $I = \displaystyle\int_0^1 \cos\left(x^2\right)dx$. Then the greatest integer less than or equal to $100I$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $I = \displaystyle\int_0^1 \cos\left(x^2\right)dx$ है। तब $100I$ से कम या उसके बराबर महत्तम पूर्णांक _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Two inequalities for $t \geq 0$, proved by monotonicity. Let $\phi(t) = \cos t - 1 + \frac{t^2}{2}$: $\phi(0) = 0$ and $\phi'(t) = t - \sin t \geq 0$ (since $\sin t \leq t$), so ϕ is increasing and $\phi(t) \geq 0$, i.e. $\cos t \geq 1 - \frac{t^2}{2}$.
Let $\psi(t) = 1 - \frac{t^2}{2} + \frac{t^4}{24} - \cos t$: $\psi(0) = 0$, $\psi'(t) = -t + \frac{t^3}{6} + \sin t$, $\psi'(0) = 0$, and $\psi''(t) = -1 + \frac{t^2}{2} + \cos t = \phi(t) \geq 0$. So ψ' is increasing, $\psi' \geq 0$, ψ is increasing and $\psi \geq 0$, i.e. $\cos t \leq 1 - \frac{t^2}{2} + \frac{t^4}{24}$.
Put $t = x^2$ $(0 \leq x \leq 1)$: $1 - \frac{x^4}{2} \leq \cos(x^2) \leq 1 - \frac{x^4}{2} + \frac{x^8}{24}$, with equality only at $x = 0$.
Integrate over $[0, 1]$: $1 - \frac{1}{10} < I < 1 - \frac{1}{10} + \frac{1}{216}$, i.e. $0.9 < I < 0.90463$.
So $90 < 100I < 90.47$, and the greatest integer less than or equal to $100I$ is 90. (Numerically $I = 0.90452\ldots$)

Final Answer: 90

Question 34

English

The value of $\displaystyle\int_0^4 x^2\sqrt{4x-x^2}\,dx$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$\displaystyle\int_0^4 x^2\sqrt{4x-x^2}\,dx$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Put $x = 4\sin^2\theta$, increasing on $[0, \frac{\pi}{2}]$, so $x : 0 \to 4$ corresponds to $\theta : 0 \to \frac{\pi}{2}$; $dx = 8\sin\theta\cos\theta\,d\theta$.

$4x - x^2 = x(4-x) = 4\sin^2\theta \cdot 4\cos^2\theta$, so $\sqrt{4x-x^2} = 4\sin\theta\cos\theta$ (both factors non-negative), and $x^2 = 16\sin^4\theta$.

The integral becomes $\int_0^{\pi/2} 16\sin^4\theta \cdot 4\sin\theta\cos\theta \cdot 8\sin\theta\cos\theta\,d\theta = 512\int_0^{\pi/2}\sin^6\theta\cos^2\theta\,d\theta$.

Wallis' formula (both exponents even, so a factor $\frac{\pi}{2}$): $\int_0^{\pi/2}\sin^6\theta\cos^2\theta\,d\theta = \dfrac{(5\cdot3\cdot1)(1)}{8\cdot6\cdot4\cdot2}\cdot\dfrac{\pi}{2} = \dfrac{15}{384}\cdot\dfrac{\pi}{2} = \dfrac{5\pi}{256}$.

So the integral is $512\cdot\dfrac{5\pi}{256} = 10\pi = 31.4159\ldots$, which is 31.42 to two decimal places.

Final Answer: 31.42

Question 35

English

For a real number $p > -1$, let $W(p) = \displaystyle\int_0^{\pi/2}\sin^p x\,dx$. Then the value of $W\left(\frac{5}{2}\right)W\left(\frac{7}{2}\right)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

वास्तविक संख्या $p > -1$ के लिए माना $W(p) = \displaystyle\int_0^{\pi/2}\sin^p x\,dx$ है। तब $W\left(\frac{5}{2}\right)W\left(\frac{7}{2}\right)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For non-integer p Wallis' product formula does not apply directly; use the Gamma form of the same integral: $\int_0^{\pi/2}\sin^p x\,dx = \dfrac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{1}{2}\right)}{2\,\Gamma\left(\frac{p}{2}+1\right)}$, with $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$.

Then $W(p)W(p+1) = \dfrac{\pi}{4}\cdot\dfrac{\Gamma\left(\frac{p+1}{2}\right)}{\Gamma\left(\frac{p}{2}+1\right)}\cdot\dfrac{\Gamma\left(\frac{p}{2}+1\right)}{\Gamma\left(\frac{p+3}{2}\right)} = \dfrac{\pi}{4}\cdot\dfrac{\Gamma\left(\frac{p+1}{2}\right)}{\Gamma\left(\frac{p+1}{2}+1\right)} = \dfrac{\pi}{4}\cdot\dfrac{2}{p+1} = \dfrac{\pi}{2(p+1)}$.

For integer $p = n$ this is the familiar Wallis consequence $W_nW_{n+1} = \dfrac{\pi}{2(n+1)}$, which follows from the Wallis reduction $W_{n+2} = \frac{n+1}{n+2}W_n$; the same reduction holds for real p, and the Gamma form supplies the starting value.

With $p = \frac{5}{2}$: $W\left(\frac{5}{2}\right) W\left(\frac{7}{2}\right) = \dfrac{\pi}{2 \cdot \frac{7}{2}} = \dfrac{\pi}{7} = 0.4487\ldots$, which is 0.45 to two decimal places.

Final Answer: 0.45

Question 36

English

For each non-negative integer n, let $I_n = \displaystyle\int_0^1 (1-x)^n e^x \, dx$. If $I_4 = ae+b$, where a and b are integers, then the value of $a + b$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

प्रत्येक अऋणात्मक पूर्णांक n के लिए माना $I_n = \displaystyle\int_0^1 (1-x)^n e^x \, dx$ है। यदि $I_4 = ae + b$ है, जहाँ a तथा b पूर्णांक हैं, तो $a + b$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For $n \geq 1$ integrate by parts with $u = (1-x)^n$, $dv = e^x dx$: $I_n = [(1-x)^n e^x]_0^1 + n\int_0^1 (1-x)^{n-1} e^x dx = (0-1) + nI_{n-1}$.

So $I_n = nI_{n-1} - 1$, with $I_0 = \int_0^1 e^x dx = e - 1$.

$I_1 = (e-1) - 1 = e - 2$; $I_2 = 2(e-2) - 1 = 2e - 5$; $I_3 = 3(2e-5) - 1 = 6e - 16$; $I_4 = 4(6e-16) - 1 = 24e - 65$.

Since e is irrational, $ae + b = 24e - 65$ with integers a, b forces $a = 24$, $b = -65$. Check: $24e - 65 = 0.2388\ldots$, which lies between $\int_0^1 (1-x)^4 dx = 0.2$ and $e \cdot 0.2$.

$a + b = 24 - 65 = -41$.

Final Answer: -41

Question 37

English

Considering the principal value of $\cos^{-1}$, for each non-negative integer n let $I_n = \int_0^1 x^n \cos^{-1} x \, dx$. If $I_6 = \frac{p}{q}$, where p and q are positive integers with $\gcd(p, q) = 1$, then the value of $p+q$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$\cos^{-1}$ के मुख्य मान को लेते हुए, प्रत्येक अऋणात्मक पूर्णांक n के लिए माना $I_n = \int_0^1 x^n \cos^{-1} x \, dx$ है। यदि $I_6 = \frac{p}{q}$ है, जहाँ p तथा q धनात्मक पूर्णांक हैं तथा $\gcd(p, q) = 1$ है, तो $p + q$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Integrate by parts with $u = \cos^{-1} x$, $dv = x^n dx$: $I_n = \left[\frac{x^{n+1}}{n+1} \cos^{-1} x\right]_0^1 + \frac{1}{n+1} \int_0^1 \frac{x^{n+1}}{\sqrt{1-x^2}} dx$. The bracket is 0 at both ends ($\cos^{-1} 1 = 0$ and $x^{n+1} = 0$ at 0).

So $I_n = \frac{1}{n+1} J_{n+1}$, where $J_m = \int_0^1 \frac{x^m}{\sqrt{1-x^2}} dx$. With $x = \sin\theta$ ($\theta : 0 \to \frac{\pi}{2}$), $J_m = \int_0^{\pi/2} \sin^m \theta \, d\theta$, a Wallis integral, so $J_m = \frac{m-1}{m} J_{m-2}$.

Reduction formula for I_n: $(n+1) I_n = J_{n+1} = \frac{n}{n+1} J_{n-1}$ and $J_{n-1} = (n-1) I_{n-2}$, so $I_n = \frac{n(n-1)}{(n+1)^2} I_{n-2}$ for $n \geq 2$.

$I_0 = \int_0^1 \cos^{-1} x \, dx = \int_0^{\pi/2} \sin\theta \, d\theta = 1$.

$I_2 = \frac{2 \cdot 1}{9} \cdot 1 = \frac{2}{9}$; $I_4 = \frac{4 \cdot 3}{25} \cdot \frac{2}{9} = \frac{8}{75}$; $I_6 = \frac{6 \cdot 5}{49} \cdot \frac{8}{75} = \frac{240}{3675} = \frac{16}{245}$.

Check by Wallis directly: $I_6 = \frac{1}{7} J_7 = \frac{1}{7} \cdot \frac{6 \cdot 4 \cdot 2}{7 \cdot 5 \cdot 3 \cdot 1} = \frac{1}{7} \cdot \frac{16}{35} = \frac{16}{245}$.

$\gcd(16, 245) = 1$, so $p + q = 16 + 245 = 261$.

Final Answer: 261